

The Supply Chain of Intelligence

A Layered Framework for Value Accrual and Defensibility in Generative-AI Software

Anand Arivukkarasu

Independent; former Product Leader, Meta (Instagram)

Correspondence: supplychainofai.com

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Abstract

This paper develops the Supply Chain of Intelligence, a descriptive framework for locating where economic value and defensibility accrue in generative-AI software. The central claim is that an AI product is not best understood as a single application or feature, but as a position across a vertical chain of ten layers: physical resources, infrastructure, data, models, gatekeeping, access, execution, orchestration, surface, and memory. The framework complements existing strategy vocabularies, the value chain, platform economics, the resource-based view, and Jobs-to-be-Done, by separating customer demand from supply-side durability. It introduces four stylized regularities: intelligence commoditizes downward; value accrues at bottlenecks; the surface captures attention while the chain captures power; and generation and verification should be separable in high-trust domains. It also identifies a recurring application-layer pattern, the Defensible Triangle, in which proprietary data, deep execution, and compounding memory reinforce one another. The paper situates the framework against two contemporaneous academic treatments, the AI Intelligence Stack of Dutta and Doz (2026) and the depth theory of Sawhney (2026), and argues that the three are complements addressing different decision-makers. The contribution is taxonomic and explanatory rather than empirical: the framework is offered as a vocabulary for product leaders, founders, and investors to reason about AI-native products, not as a tested causal model or a valuation method.

Keywords: generative AI; software strategy; value chain; defensibility; moats; platform economics; foundation models; agentic systems; intelligence stack.

JEL classification (suggested): L86; O33; L10; M13; M15.

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<https://supplychainofai.com/paper>

1. Introduction: why AI needs a supply-chain map

Every wave of software produces products that look, briefly, like durable companies. The generative-AI wave has compressed that cycle. A product launches on top of a frontier model, attracts usage and attention, and may then be absorbed when a model provider, cloud platform, operating-system owner, or incumbent workflow provider ships the same capability as a bundled feature. Practitioners summarize the pattern with a crude phrase: wrappers become features. The phrase is useful, but it is not precise enough to guide product strategy, investment judgment, or roadmap design.

This paper offers a more precise map. Its core claim is that a generative-AI product is a position in a supply chain, not a point on a feature list. Value moves through that chain in stages: scarce physical resources feed compute; compute supports data and models; models pass through gates of trust, then access, execution, orchestration, surface, and memory. At each stage, a different actor may capture value. Many AI products occupy one visible layer, often the surface. Durable products either own a scarce layer deeply or combine several layers that reinforce one another.

The framework is intentionally modest in method. It is a conceptual taxonomy, not an empirical study. It does not claim that layer ownership mechanically causes survival, or that every product can be scored with precision. Its purpose is to help practitioners name structural positions otherwise discussed informally: which layer does a company actually own; which layer is it renting; which layer can absorb it; which adjacent layer could it move into; and where does durable value sit when general model capability becomes widely available?

1.1 Intended use

The framework is designed for three related decisions:

- Product leaders use it to identify whether a roadmap strengthens a durable layer or merely polishes an absorbable surface.
- Founders use it to ask which layer must be entered before the current layer commoditizes.
- Investors use it to distinguish a workflow with compounding advantage from a thin interface that may be bundled away.

One strategic question sits under all three: when intelligence becomes cheaper and more widely distributed, which layer stays scarce enough to support durable value? Put as a decision, the question is which layers a company should own, which it can safely rent, and which it must expect to be absorbed into. Figure 1 presents the full chain.

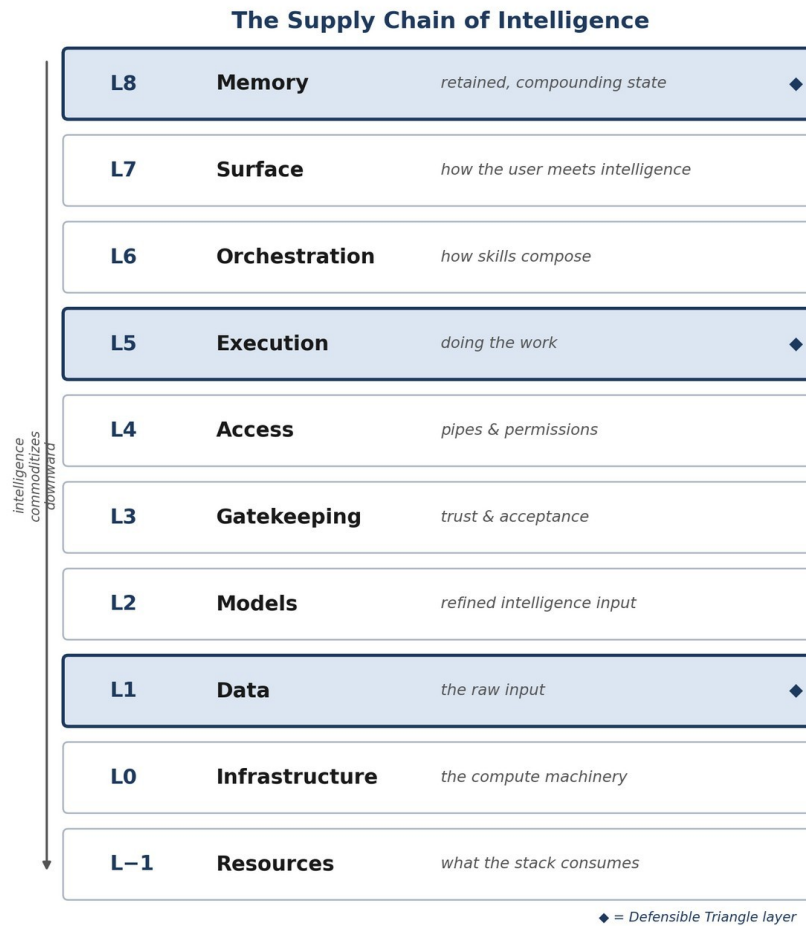


Figure 1. The Supply Chain of Intelligence: ten layers, top to bottom. Diamonds mark the three Defensible-Triangle layers (Section 5).

2. Relationship to existing strategy frameworks

The Supply Chain of Intelligence does not replace established strategy tools. It adapts them to a software stack in which a general-purpose capability, model-based intelligence, is supplied by a small number of foundation-model providers and diffuses through APIs, copilots, operating systems, and enterprise workflows. Existing vocabularies each capture part of this shift and miss part.

2.1 Value-chain logic

Porter's value chain decomposes the firm into activities that create margin. The present framework is value-chain-like, but its unit of analysis is the industry stack rather than the internal firm; its layers are technical and organizational strata rather than business functions such as logistics or service (Porter, 1985).

2.2 Resource-based and capability views

The resource-based view asks whether a firm controls assets that are valuable, rare, hard to imitate, and organizationally embedded (Barney, 1991). Dierickx and Cool (1989) add that such assets are stocks accumulated over time, which is why they resist imitation even by better-capitalized rivals. The framework

translates that question into stack position: proprietary outcome data, regulatory acceptance, execution playbooks, and customer-specific memory are accumulated stocks tied to specific layers. This connects to the broader question of who profits from innovation when complementary assets and control points sit outside the innovator (Teece, 1986).

2.3 Platform economics

Platform economics explains why some layers capture power: they intermediate access, aggregate demand, set rules, and bundle adjacent capabilities (Rochet and Tirole, 2003; Eisenmann, Parker, and Van Alstyne, 2006). The recurring drama in AI is that a lower or broader platform layer absorbs a thinner layer above it. Platform theory explains the predator; the layered map helps identify the prey's structural options.

2.4 Jobs-to-be-Done

Jobs-to-be-Done clarifies what a customer is trying to accomplish (Christensen et al., 2016). It is strong on demand-side clarity and weak on supply-side defensibility. Two products can serve the same job and meet opposite outcomes because one owns a scarce layer while the other depends on a commoditizing capability. A compact statement of the complement: **Jobs-to-be-Done locates demand; the Supply Chain of Intelligence locates defensibility.**

2.5 AI economics and foundation models

Recent AI systems behave like broad capability substrates rather than narrow software modules. Foundation models can be adapted across tasks and distributed through APIs, embedded interfaces, and agentic workflows (Brown et al., 2020; Bommasani et al., 2021). This changes where advantage lives: the model may be necessary but is often not sufficient. Advantage instead accrues in data rights, trust gates, workflow control, evaluation, orchestration, or memory (Agrawal, Gans, and Goldfarb, 2018; Jacobides, Cennamo, and Gawer, 2018). Jacobides, Brusoni, and Candelon (2021) map the AI ecosystem into enablement, production, and consumption roles and show that architectural position, not adoption level, shapes competitive dynamics; the present framework can be read as a finer-grained decomposition of that architecture, with ten layers in place of three roles. Iansiti and Lakhani (2020) describe the learning loops through which AI-centred operating models compound; Section 5 locates those loops at specific layers.

2.6 Contemporaneous layered treatments of AI strategy

Two academic working papers released after the framework's public publication develop closely related arguments, and the relationship deserves to be stated explicitly.

Dutta and Doz (2026) introduce the AI Intelligence Stack, a five-layer architecture of Data, Model, Decision, Routing, and Interface, and recast the three canonical strategy questions, where to compete, how to win, and what capabilities to build, through it. Their central positioning claim is that firms must decide which layers to control, which to access through partners, and which to accept as structural dependencies; their central learning claim is that advantage compounds through recursive feedback across layers. The frameworks correspond closely at their core: their Data, Model, Decision, Routing, and Interface layers map onto L1 Data, L2 Models, L5 Execution, L6 Orchestration, and L7 Surface respectively, and their

Routing layer, which they also call the workflow orchestration layer, is functionally the closest match. The differences follow from audience. Dutta and Doz state that their framework applies most directly to firms competing as users rather than builders of AI, and their examples are incumbents adopting intelligence. The Supply Chain of Intelligence was built from the AI industry's own supply chain and from the fate of AI-native product companies, which is why it extends below data into infrastructure and physical resources, treats gatekeeping and access as layers in their own right, and places memory above the interface as a distinct strategic asset. Read together, the two frameworks are complements: one for the enterprise adopting intelligence, one for the companies supplying it.

Sawhney (2026) develops a theory of depth, defined as the stock of co-specialized complementary assets in four classes, data capital, encoded expertise, workflow embedment, and trust assets, that converts rented model intelligence into defensible value, and operationalizes it as a seven-dimension diagnostic. His four classes correspond to positions in the present chain: data capital to proprietary and outcome data (L1), encoded expertise to execution (L5), workflow embedment to execution and orchestration (L5 and L6), and trust assets to gatekeeping (L3). His flywheel, in which embedded usage generates outcome data that refines models and playbooks, is the Defensible Triangle of Section 5 expressed as dynamics rather than structure. Sawhney cites the framework as the most systematic practitioner treatment of where value accrues and takes up Law IV directly; his contribution is to supply the formal propositions and falsification conditions that a practitioner taxonomy lacks. Table 3 summarizes the correspondence.

Concept	Supply Chain of Intelligence (this paper)	Dutta and Doz (2026)	Sawhney (2026)
Unit of analysis	The industry supply chain of intelligence; AI-native product companies	The firm's internal intelligence architecture; AI-adopting enterprises	The application firm above a rented substrate
Layers or classes	Ten layers, fifty sublayers (L-1 to L8)	Five layers: Data, Model, Decision, Routing, Interface	Four asset classes: data capital, encoded expertise, workflow embedment, trust assets
Core strategic question	Which layer do you own, rent, or get absorbed into?	Which layers to control, partner, or accept dependence upon?	How deep is the stock of co-specialized assets above the substrate?
Mechanism of advantage	Bottleneck scarcity; the Defensible Triangle (L1, L5, L8)	Learning velocity across recursive layers; architectural integration	Accumulation, flywheel, trust asymmetry, renewal
Where the frameworks meet	L1, L2, L5, L6, L7 and the own-or-rent question	Data, Model, Decision, Routing, Interface	D1 to L1; D2 to L5; D3 to L5/L6; D4 to L3
Distinctive to this paper	Resources, Infrastructure, Gatekeeping, Access, Memory as layers; Law IV; the three currents	Recasting the three classical strategy questions; human-AI capability at each layer	Six falsifiable propositions; stranding of assets by a moving substrate; the Depth Diagnostic

Table 3. Correspondence between the Supply Chain of Intelligence and two contemporaneous academic frameworks.

3. The ten layers

The framework numbers layers from L-1 at the physical base to L8 at the top, ten layers in all. The numbering keeps the model layer low in the stack, reflecting the argument that models increasingly behave like refined inputs to products rather than the entirety of the product. Table 1 summarizes the layers; Appendix A decomposes each into five sublayers, fifty in total.

Layer	Name	What it is
L8	Memory	What the system retains and compounds: session state, user and entity profiles, outcome histories, institutional knowledge, and learned context.
L7	Surface	How the user meets the intelligence: conversational, visual, embedded, transactional, and ambient interfaces.
L6	Orchestration	How skills compose into outcomes: agent loops, role routing, context and state management, human-in-the-loop design, runtime assurance.
L5	Execution	The work itself: domain skills, decision frameworks, retrieval-augmented workflows, operating playbooks, task-specific procedures.
L4	Access	The pipes and permissions: APIs, protocols, access governance, real-time infrastructure, agent identity and provenance.
L3	Gatekeeping	Trust and acceptance: compliance, quality, safety and security, editorial review, and distribution gates.
L2	Models	Refined intelligence inputs: foundation, specialized, embedding and retrieval, routing, and reasoning models.
L1	Data	The raw and structured input: public, proprietary, behavioral, outcome, and synthetic data; licensed corpora.
L0	Infrastructure	Compute machinery: silicon, data centers, interconnect fabric, cloud compute, edge and on-device compute.
L-1	Resources	What the stack consumes: energy, thermal and water management, fabrication capacity, critical materials, skilled trades.

Table 1. The ten layers of the generative-AI supply chain, top to bottom.

Three design choices are central. First, memory sits at the top. As model capability diffuses, the durable asset often migrates toward what the system accumulates and cannot re-derive from a public model: customer state, outcome histories, institutional preferences, operational context. Second, gatekeeping is a first-class layer. Where outputs carry regulatory, fiduciary, safety, or reputational weight, the right to be trusted is itself a scarce position. Third, the chain runs below the model into infrastructure and physical resources, because in periods of compute scarcity those layers set the price of every layer above them and can hold value that no application-layer position can reach.

4. Four structural regularities

Four regularities follow from the layered view. They are called laws for mnemonic clarity, but should be read as stylized mechanisms rather than scientific laws. Each can be resisted by distribution, capital, timing, regulation, or exceptional execution.

Law I. Intelligence commoditizes downward

A product whose only differentiator is general model capability is exposed to absorption by the layer below it. When the claim is only that the model can perform a generic task, and the provider can ship that task broadly, the product's distinct value narrows. Distribution and switching costs delay this; they do not remove the underlying pressure.

Law II. Value accrues at bottlenecks

Durable margin tends to sit where the chain is scarce and hard to replicate: proprietary data, regulatory acceptance, distribution control, domain execution, compounding memory. The strategic question is not only what the product does, but which scarce layer it controls and whether that scarcity will persist.

Law III. The surface captures attention; the chain captures power

A polished interface wins usage, narrative, and early growth, yet a surface unconnected to a deeper control point is vulnerable. Enduring products pair a usable surface with ownership of another layer: data, gatekeeping, execution, orchestration, access, or memory. Attention can be rented; structural power must be owned or contractually protected.

Law IV. Generation and verification should be separable

Where outputs carry fiduciary, regulatory, safety, or reputational weight, the actor that generates a result and the actor that verifies it often need to be separable; a generator certifying its own high-stakes output creates a trust problem. This is why gatekeeping can become an independent value-bearing layer rather than a feature of the generator. The law is partly descriptive and partly normative: it describes where value may accrue and where trust should sit if the system is to be credible. A corollary is that platform providers expand into every adjacent layer of value but an independent gatekeeping position above the generator is structurally permanent in regulated domains, which is why verification positions such as Vanta above AWS or Snyk above Copilot have so far survived platform envelopment.

4.5 Three horizontal currents

The layers describe vertical structure. Three horizontal forces move value across the chain regardless of where a firm sits, and any reading of a position should account for them. Demand gravity is the location of budget: defensibility without a budgeted buyer is worth nothing, and as the model layer commoditizes, budget migrates toward the layers that deliver outcomes. Attention economics is the scarcity created by abundant generation: when output is cheap, default placement and ownership of the on-ramp command rent, which is why operating-system and productivity-suite owners act as landlords of the surface layer. Capital flows are the reflexive distortion of layer valuations by funding, which overheats fashionable layers and starves the bottlenecks that Law II says will eventually hold value.

5. The Defensible Triangle

For application-layer firms that do not own compute or foundation models, a recurring durable pattern combines three reinforcing layers (Figure 2):

1. **Proprietary data (L1).** Inputs, rights, or observed behavior the model layer cannot easily acquire.

2. **Deep execution (L5).** Domain skills, decision procedures, and operating playbooks encoded into the workflow.
3. **Compounding memory (L8).** Customer-specific or network-level state that improves with use and raises the cost of leaving.

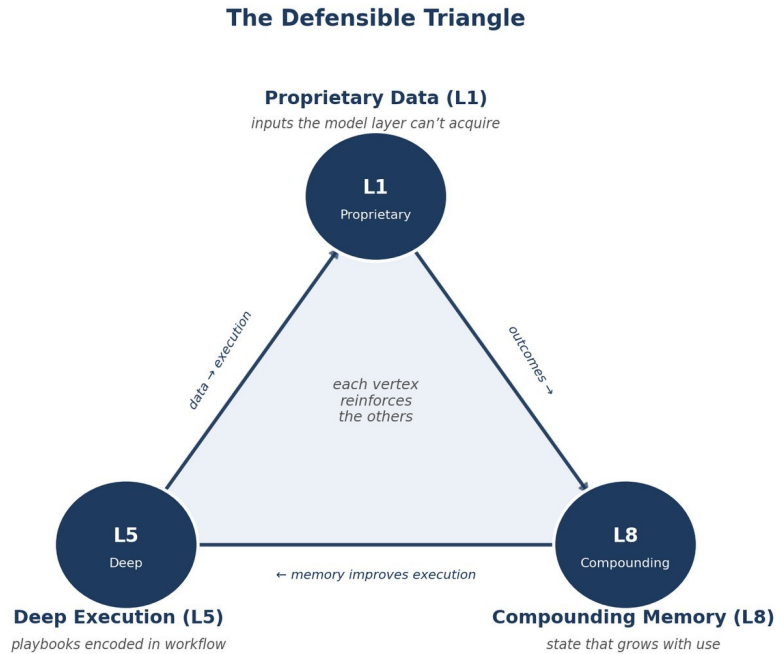


Figure 2. The Defensible Triangle. Data improves execution; execution generates outcome data; outcomes compound into memory; memory improves future execution.

A competitor may replicate a visible feature, but cannot instantly reproduce a customer's accumulated state, the workflow-specific judgments embedded in execution, or the historical feedback loop connecting them. The triangle is not the only route to defensibility: a firm may also survive by owning one scarce layer with unusual depth, such as infrastructure control, licensed data, regulatory access, or a trusted distribution gate. The triangle is a pattern for application-layer firms, not a universal rule.

6. Illustrative applications

The cases below are illustrative archetypes, not a validated sample. They were selected because they clarify the categories, and the trajectory descriptions are consistent with public reporting as of early 2026 rather than independently audited. Each links to a longer case note in the author's public corpus at supplychainofai.com/analysis. A later empirical version should replace these with coded cases, explicit data sources, and falsifiable predictions (Section 7.5). The point here is to show how the vocabulary changes the diagnosis.

6.1 Same job, different structural positions: writing and coding assistance

Several products serve the same broad job, help a user write or code, while occupying different layers. A standalone copy-generation tool (Jasper) sat largely on the surface (L7) and saw its differentiation compress when model providers shipped equivalent generation broadly, consistent with Law I. A grammar-and-style assistant (Grammarly) endured longer by owning distribution into browsers and editors (L4 and L7), until a larger owner of that same distribution layer (Microsoft, via the office suite) integrated a model directly. A coding assistant built into the editor (Cursor) fared differently again by accumulating workflow control, an indexing pipeline, an agent loop, and project memory (L4, L6, and L8), several reinforcing layers rather than one. Jobs-to-be-Done groups these products together; the layer map separates their exposure.

Corpus notes: supplychainofai.com/analysis/jasper-vs-grammarly-copilot; supplychainofai.com/analysis/cursor-ide-consolidation

6.2 Data mispackaged as surface

Some communities accumulated valuable human knowledge but presented it as surface content (L7). When that content became training and retrieval input for model-layer systems (L2), an L1 data asset was captured economically by a layer owned by someone else, while the community that produced it captured little, a pattern visible in the post-2022 traffic decline of large developer-knowledge sites such as Stack Overflow. The framework names the error: a value-bearing input was treated as content rather than protected as a scarce data layer.

Corpus note: supplychainofai.com/analysis/stack-overflow-decline

6.3 Memory as the moat in enterprise agents

Two enterprise customer-service agents can look identical in a demonstration while differing structurally. One (Sierra) was architected from inception around behavioral data (L1), operating playbooks (L5), and per-customer compounding memory (L8), the full triangle, where every resolution updates a customer-specific model. The other bolts an execution layer onto an existing data platform with no compounding loop. The first compounds; the second performs. The difference determines whether each use makes the next use more defensible.

Corpus note: supplychainofai.com/analysis/sierra-vs-salesforce

6.4 Single-layer fortresses

Not every durable position requires a triangle. A supplier of unusually scarce compute infrastructure (NVIDIA) defends an L0 position. A provider of legally cleared, licensed data (Adobe Firefly) defends an L1 and L3 position: the gate, not the generator, is the moat. A vertical data-plus-gate stack in a regulated domain (Bloomberg) defends L1, L2, and L3. These cases discipline the framework against over-claiming the triangle: depth at one scarce layer, often far below the visible surface, is a second valid route to durability.

Corpus notes: supplychainofai.com/analysis/adobe-firefly-licensed-data; supplychainofai.com/analysis/bloomberg-gpt-vertical-fortress

Table 2 collects these readings. The layer assignments are the author's structural judgment, not measurements; the same product could be placed slightly differently by another analyst (Section 7.2).

Company	Category	Layer position	Absorption risk	Framework reading
Jasper	AI writing	Surface only (L7)	High	Surface exposure; absorbed when generation went free
Grammarly	Writing assistant	Distribution and surface (L4, L7)	Medium	Protected by distribution until a bigger L4 owner integrated
Cursor	Coding	Access, orchestration, memory (L4, L6, L8)	Lower	Reinforcing layers; the model is the commodity
Stack Overflow	Developer knowledge	Data mispackaged as content (L1 to L7)	High	Value-bearing input captured by the ingesting layer
Sierra	Enterprise agents	Data, execution, memory (L1, L5, L8)	Lower	Full Defensible Triangle; compounds with use
NVIDIA	AI compute	Infrastructure (L0)	Low to medium	Single-layer fortress
Adobe Firefly	AI images	Licensed data and gate (L1, L3)	Medium to low	Gatekeeping moat, not the generator

Table 2. Illustrative archetypes. Layer assignments are structural judgment; trajectories are consistent with public reporting, not audited.

7. Limits and research agenda

A useful framework must name where it is weak. The Supply Chain of Intelligence is not a substitute for evidence, product taste, market timing, capital access, or execution quality.

7.1 Taxonomy, not causal proof

The framework classifies positions and proposes mechanisms; it does not prove that layer position causes survival, valuation, or margin. Because it is descriptive, no single case can falsify it the way a prediction could, a feature for a vocabulary and a liability for anyone tempted to treat it as a predictive model. It is not one. Sawhney (2026) shows what the next step looks like: propositions with stated falsification conditions.

7.2 Fuzzy boundaries

Layer assignment is partly judgment-based: a capability may sit between execution and orchestration; a data asset may become memory once connected to outcomes; a distribution gate may function as both access and gatekeeping. The framework reduces ambiguity relative to having no map, but any numeric score built on it, including the audit in Appendix B, should be treated as structured judgment, not measurement.

7.3 Moving bottlenecks

The layers are more stable than the scarcity of any particular layer. Synthetic data may relieve some data bottlenecks; model routing may commoditize parts of orchestration; regulation may create or erase

gatekeeping positions; memory infrastructure may become easier to buy. A map drawn in 2026 should be redrawn as bottlenecks move.

7.4 Timing, capital, and execution

A company can occupy a strong position and still fail; another can occupy a weak position and survive on distribution, speed, capital, or execution. The framework locates where durable value can sit; it is largely silent on when and on who executes well, both of which often dominate outcomes.

7.5 Empirical agenda

The next version should become testable. One path is to code a population of AI-native products by primary layer, adjacent-layer ownership, data rights, trust gates, workflow depth, memory depth, and distribution control; outcomes could include retention, margin, acquisition, feature absorption, enterprise adoption, or valuation change, with controls for funding, distribution, team quality, incumbent response, and timing. A second path is to test the correspondence claims of Section 2.6: if the frameworks are complements, a firm's position in the Supply Chain of Intelligence should predict which of Dutta and Doz's positioning choices it faces, and its Defensible Triangle should predict its depth profile under Sawhney's diagnostic. The framework would then move from vocabulary to a set of hypotheses about product durability.

8. Conclusion

The Supply Chain of Intelligence proposes a simple reframing: an AI product is a position in a layered chain. The visible feature is rarely the whole product. Durable value tends to sit where the chain is scarce, trusted, embedded, or compounding. The four laws and the Defensible Triangle are devices for making those positions easier to see and discuss.

The framework should be judged by whether it improves decisions. Does it help a product leader see that a roadmap is overinvested in surface and underinvested in memory? Does it help a founder identify which layer must be owned before the current one commoditizes? Does it help an investor distinguish a thin interface from a workflow with accumulating advantage? If so, it earns practical value before it becomes an empirical theory. If not, the agenda in Section 7.5 shows where it must be sharpened.

Notes on method and attribution

This is a conceptual working paper. It reports no original dataset and conducts no statistical test. The framework, the four laws, and the Defensible Triangle are the author's synthesis. The framework was first published publicly on May 23, 2026, on LinkedIn and at supplychainofai.com; the first version of this paper followed in May 2026, and this version adds Section 2.6, Section 4.5, Appendix B, and corrections. The public set of related diagrams and case notes is maintained at supplychainofai.com and may be cited or extended with attribution. Any future version making claims about specific companies, valuations, or performance trajectories should cite primary sources or clearly label the claims as illustrative.

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Appendix A. The fifty sublayers

Each layer decomposes into five sublayers. The framework marks sublayers that tend to be structurally defensible, those hardest for an adjacent layer to absorb, though, per Section 7.2, the boundaries are judgment-based.

Layer	Five sublayers
L8 Memory	Session and short-term memory; user and entity profiles; aggregated network learning; institutional knowledge; learned world models
L7 Surface	Conversational; visual interfaces and media; embedded and copilot; transaction surface; async and ambient surfaces
L6 Orchestration	Agent loops; human-in-the-loop; role routing and task decomposition; context and state management; runtime assurance and learning loops
L5 Execution	Domain execution; decision frameworks and reasoning scaffolds; retrieval-augmented workflows; operating playbooks; interactional skills
L4 Access	API and integration layer; agent interface protocols; access governance; real-time interaction infrastructure; agent identity and provenance
L3 Gatekeeping	Compliance gates; quality gates; safety and security; editorial gates; distribution gates
L2 Models	Foundation models; specialized models; embedding and retrieval; model routing and composition; reasoning models
L1 Data	Public and open data; proprietary data; behavioral data; outcome data; synthetic data
L0 Infrastructure	Silicon; data centers; interconnect fabric; compute access and cloud; edge and on-device compute
L-1 Resources	Energy; thermal and water management; fabrication and foundry; critical materials; skilled trades and human capital

Table A1. Fifty sublayers across the ten layers. Full descriptions are maintained at supplychainofai.com/framework.

Appendix B. The AI Defensibility Audit

The audit converts the framework into a structured self-assessment for a single product. Eight dimensions are each scored from 1 (fully exposed) to 5 (structurally defended). Per Section 7.2, the result is structured judgment rather than measurement, and it is most useful as a prioritization device: the lowest scores identify which layer to build next. The interactive successor to the audit, the Defensibility Map, plots a company across all fifty cells of the chain and is maintained at supplychainofai.com.

#	Dimension	Layer and what is being scored
1	Model dependency	L2. Exposure to repricing or substitution of the underlying model; a high score means product quality survives a model switch.
2	Data ownership	L1b. Proprietary data acquisition; a high score means outcome-linked data the model layer cannot acquire.
3	Workflow depth	L5. Organizational integration of execution; a high score means playbooks and decision frameworks are encoded into how the work is done.

#	Dimension	Layer and what is being scored
4	Trust gate	L3. Regulatory or institutional non-routability; a high score means acceptance a generator cannot self-supply.
5	Distribution	L4 and L3e. Ownership versus rental of the channel and on-ramp; a high score means users arrive through a gate the firm controls.
6	Memory	L8. Per-customer or network compounding; a high score means state that improves with use and cannot be re-derived from a public model.
7	Switching cost	Buyer exit friction across L5, L6 and L8; a high score means removal disrupts the customer's operations.
8	Platform exposure	L2 and L7. Host commoditization risk; a high score means the layer below has neither the means nor the incentive to bundle the product's core function.

Table B1. The eight audit dimensions. Total scores map to six bands, from Thin Wrapper through Useful Tool, Weak Moat, Workflow Product and Defensible AI System to Intelligence Gate.